



An integrated indexical and multivariate assessment of surface water quality within the Nag river, Maharashtra, India

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Abstract

The Nag River, flowing through the highly urbanized core of Nagpur, Maharashtra, serves as the primary drainage system for the city and is critically polluted due to rapid urban development and uncontrolled industrial discharges. While conventional physicochemical assessments exist, they fail to provide the quantitative, spatially resolved source apportionment necessary for targeted remediation. This study integrates WQI with PCA and CA to provide a structured, data-driven assessment of pollution sources along the Nag River, specifically Principal Component Analysis (PCA) and Cluster Analysis (CA) within a spatial framework to provide the first systematic differentiation of pollution sources (e.g. municipal sewage vs. specialized industrial effluent) and link them directly to specific land-use zones along the river's 17 km urban corridor. The aim was to holistically assess the surface water quality and quantitatively identify, map, and attribute pollution sources along this critical stretch. Nine (9) surface water samples (S1-S9) were systematically collected during the pre-monsoon season (February 2023), covering segments influenced by diverse residential, commercial, and industrial land use. Twenty physicochemical and biological parameters were analyzed, and the reliability of the hydrochemical data was confirmed using the Ionic Balance Error (IBE) validation. WQI values ranged severely from 47.05 (Good) at the upstream baseline (S1) to a maximum of 6440.38 (Unfit for all practical uses) at Yashwant Stadium (S5), confirming chronic heavy pollution. This degradation is primarily attributed to untreated municipal sewage, as indicated by extreme BOD levels up to 216.28 mg/l and non-compliant specialized industrial discharges. PCA identified three primary Varifactors (VFs) explaining 87.935% of the total variance. Varifactor 1 (44.088%) confirmed the overwhelming dominance of untreated municipal sewage (organic load, total dissolved solids, and microbiological parameters). Varifactor 2 (16.666%) was strongly associated with specialized heavy metals (Nickel and Cadmium), indicating a distinct point source industrial effluent. CA successfully categorized sampling sites into four spatial pollution clusters (C1-C4), enabling the identification of high-priority pollution hotspots that correlate directly with land use. The present study integrates the WQI-PCA-CA approach, combined with land use assessment, to provide critical insights to support evidence-based river restoration and sustainable watershed management planning.

Keywords Cluster analysis · Nag river · Principal component analysis · Water quality · Water quality index.

Introduction

Water quality plays a fundamental role in maintaining ecological balance, supporting sustainable development, and safeguarding human health. It directly influences biodiversity, agricultural productivity, industrial growth, and the overall well-being of societies. However, in recent decades, the deterioration of water resources, especially surface waters such as rivers, lakes, and reservoirs has emerged as a critical environmental concern worldwide. The rapid pace of population growth, urbanization, industrialization, and agricultural intensification has dramatically increased the

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pollutant load entering aquatic ecosystems. This escalating pressure has resulted in the depletion of clean freshwater resources, posing major challenges for developing nations struggling to balance economic growth with environmental conservation (Fathi et al. 2018; Kabir et al. 2020; Egbueri et al. 2024).

Rivers, in particular, play a vital role as lifelines of civilizations, serving as primary sources of domestic water supply, agricultural irrigation, fisheries, recreation, and industrial activities. Yet, they are also the most vulnerable to pollution, receiving a wide range of contaminants from both point and non-point sources. Anthropogenic inputs such as sewage discharge, industrial effluents, and agricultural runoff release organic matter, nutrients, heavy metals, and other pollutants that degrade water quality (Mishra et al. 2017; Zhang et al. 2023). In addition, natural processes such as soil erosion, rock weathering, and leaching of minerals further alter the aquatic chemical balance. Together, these phenomena lead to increased turbidity, changes in pH, elevated biochemical oxygen demand (BOD), and abnormal concentrations of ions and trace metals. The resulting deterioration not only disrupts aquatic ecosystems but also increases health risks for local populations dependent on river water for daily use (Chen et al. 2022; Kumar et al. 2020). Assessing river water quality is inherently complex due to the dynamic interplay of physical, chemical, and biological parameters that vary spatially and temporally (Gaury et al. 2018; Ustaoglu et al. 2021). Traditional monitoring methods generate extensive datasets that are often difficult to interpret. Large volumes of parameter measurements may obscure rather than clarify patterns of pollution, hindering efforts to implement effective water management policies. To address this challenge, scientists have developed integrated assessment techniques that transform complex datasets into easily interpretable indicators. Among these, the Water Quality Index (WQI) has been widely adopted as an efficient tool for summarizing multiple water quality parameters into a single dimensionless score that reflects the overall status of the river system (Alexakis et al. 2016; Bora and Goswami 2017; Igwe and Omeka 2022).

The WQI has become one of the most versatile and widely used tools in hydroenvironmental assessment. It translates complex chemical and physical data into a single index value, allowing easy comparison of water quality across locations and time. The usefulness of WQI lies in its capacity to simplify communication of scientific data to decision-makers, policy planners, and the general public. However, while WQI provides a holistic overview, it cannot independently pinpoint specific sources of contamination or reveal the underlying patterns driving water quality variability (Ogwueleka 2015; Mutlu and Uncumusaoglu 2022). As a result, researchers increasingly combine WQI

with advanced statistical and geospatial methods to extract deeper insights into pollution sources and distribution dynamics (Singh et al. 2022).

Multivariate statistical techniques particularly Principal Component Analysis (PCA) and Cluster Analysis (CA) have become valuable tools for identifying key pollutants and classifying sampling locations based on similar hydrochemical characteristics. PCA helps reduce large datasets to a smaller number of principal components that explain most of the variance, revealing dominant factors influencing water quality, such as industrial discharge, agricultural runoff, or sewage load (Arora and Keshri 2021; Gyimah et al. 2021). Cluster Analysis, on the other hand, groups sampling sites with comparable water quality profiles, facilitating spatial interpretation of pollution patterns and aiding in the delineation of water management zones (Mishra et al. 2018; Jehan et al. 2020). Together, PCA and CA help to overcome the limitations of WQI by linking analytical results to probable pollution sources, thereby making interpretation more meaningful for environmental management.

In recent years, integrating WQI with PCA and CA has become a robust and data-driven framework for water quality assessment worldwide. When coupled with Geographic Information Systems (GIS), this combination offers additional spatial visualization and modeling capabilities that provide a holistic understanding of river pollution. Geospatial technologies enable the mapping of spatial gradients, identification of pollution hotspots, and visualization of the influence of land use on water quality (Unigwe et al. 2022; Xu and Zhang 2023; Aghamelu et al. 2023; Omeka et al. 2024). Studies applying such combined WQI-PCA-CA-GIS approaches to major river systems such as the Ganga in India, the Danube in Europe, the Nile in Africa, and the Mekong in Southeast Asia have demonstrated how these methods can reveal complex trends in pollution variation across diverse climatic and land-use settings (Dimri et al. 2021; Panagopoulos et al. 2022; Gad et al. 2023; Tran et al. 2023). These studies not only enhanced understanding of spatial and temporal variability in water quality but also contributed valuable information for policy formulation and restoration planning.

Despite such advances, research specifically linking industrial clusters to the spatial variation of WQI and pollution patterns remains limited, particularly in densely populated urban areas of developing nations such as India. Indian rivers are often subjected to uncontrolled waste discharge due to inadequate wastewater treatment infrastructure, resulting in alarming declines in water quality. The rapid expansion of urban settlements, combined with the proliferation of small and medium-scale industrial units, has intensified pollution pressure on river systems. Urban rivers thus exhibit strong spatial variability in hydrochemical

properties, reflecting the combined impacts of domestic sewage, industrial activities, and runoff from impervious surfaces. In this context, an integrated analytical approach becomes essential to unravel the relationships between human-induced activities and water quality degradation (Chowdhury et al. 2024; Khan et al. 2023).

The Nag River in Nagpur, Maharashtra, provides a representative case of urban water pollution in India. Historically a freshwater stream supporting the city's water supply, the Nag River has been transformed into a highly polluted drainage channel due to continuous discharge of untreated domestic sewage and industrial effluents. The river traverses densely populated and industrially active zones, receiving wastewater from various sources, including textile dyeing, electroplating, small-scale metal industries, and urban runoff. Previous studies (Patil et al. 2017; Dutta et al. 2018; Jain et al. 2022) have primarily focused on individual physico-chemical parameters such as pH, BOD, COD, hardness, and dissolved oxygen to assess pollution levels. However, these isolated assessments fail to capture the integrated interactions among parameters or to link specific anthropogenic activities with spatial differences in water quality along the river stretch. As a result, policymakers and local authorities lack a unified framework for identifying pollution sources and prioritizing remediation actions.

This study aims to bridge that gap by applying an integrated WQI-PCA-CA approach to systematically evaluate the hydrochemical characteristics and spatial pollution dynamics of the Nag River. The specific objectives are threefold: (1) to assess the river's suitability for domestic, agricultural, and other utilitarian purposes based on WQI classification; (2) to identify major factors and key variables influencing water quality through Principal Component Analysis; and (3) to categorize sampling sites into distinctive clusters using Cluster Analysis to distinguish highly polluted zones from relatively cleaner segments. The integration of these methods provides a comprehensive, data-driven framework that facilitates better interpretation of pollution sources, their spatial distribution, and relative significance.

Beyond its local importance, the study holds broader scientific and policy relevance. The integrated WQI-PCA-CA framework developed here can serve as a replicable model for assessing urban river systems across developing regions facing similar industrial and demographic pressures. It also offers essential guidance for designing targeted restoration programs, optimizing water management strategies, and prioritizing investments in wastewater infrastructure. Furthermore, the methodology aligns with global initiatives such as the United Nations Sustainable Development Goal 6 (SDG 6), which advocates ensuring the availability and sustainable management of clean water and sanitation for

all. By advancing methodological integration and providing empirical evidence from a critical urban river system, this research contributes to both local and global efforts to safeguard water resources through scientific, data-supported approaches.

Materials and methods

Study area

The Nag River passes through Nagpur City, located in Maharashtra, India. The monitored segment covers the entire 17-kilometer urban stretch, originating from lava hills near Wadi and flowing to the confluence point. The river's depth ranges from 2 to 4.5 m, while its width ranges from 12 to 40 m (Nag River Basin Action Plan 2011).

Nagpur has a tropical wet and dry climate with three seasons: summer, monsoon, and winter. The mean annual temperature of the region is approximately 27 °C, with summers exceeding 40 °C and winters being moderate, with temperatures sometimes going below 10 °C. The city gets an average annual rainfall of around 1,200 mm, mostly from the southwest monsoon from June to September. The humidity percentage varies throughout the year, from 20% during dry months to more than 80% during the height of monsoon months. The soil cover in the Nag River area is predominantly black cotton soil, derived from the underlying Deccan Trap basalt (temperature, total rainfall, humidity and soil type, https://mausam.imd.gov.in/responsive/rainfall_statistics.php). This lithological context is vital because the natural weathering of the basalt influences the baseline concentration of major ions (such as Ca^{2+} , Mg^{2+} , and SO_4^{2-}). Understanding this geological baseline is crucial for accurately distinguishing natural hydrochemical variations from the highly significant anthropogenic contamination observed along the river's course. The basin has a high urban population density, with Nagpur being a megacity in central India with a population size of over 11,000 individuals per square kilometer (Nag River Basin Action Plan 2011). The process of fast growth and industrialization has resulted in higher contamination loads in the river. The city is home to some big industries, such as power generation, textile, automobile, and pharmaceutical units, which all contribute to wastewater effluent (Muller et al. 2020) (Fig. 1).

Water samples from nine (9) stations, systematically labeled and coordinates for S1 to S9 shown in Fig. 2; Table 1, respectively, were selected based on major environmental, ecological, and geographical factors to assess various physicochemical parameters of the Nag River. The site selection was systematic, guided by proximity to major human interventions, including known sources of pollution,

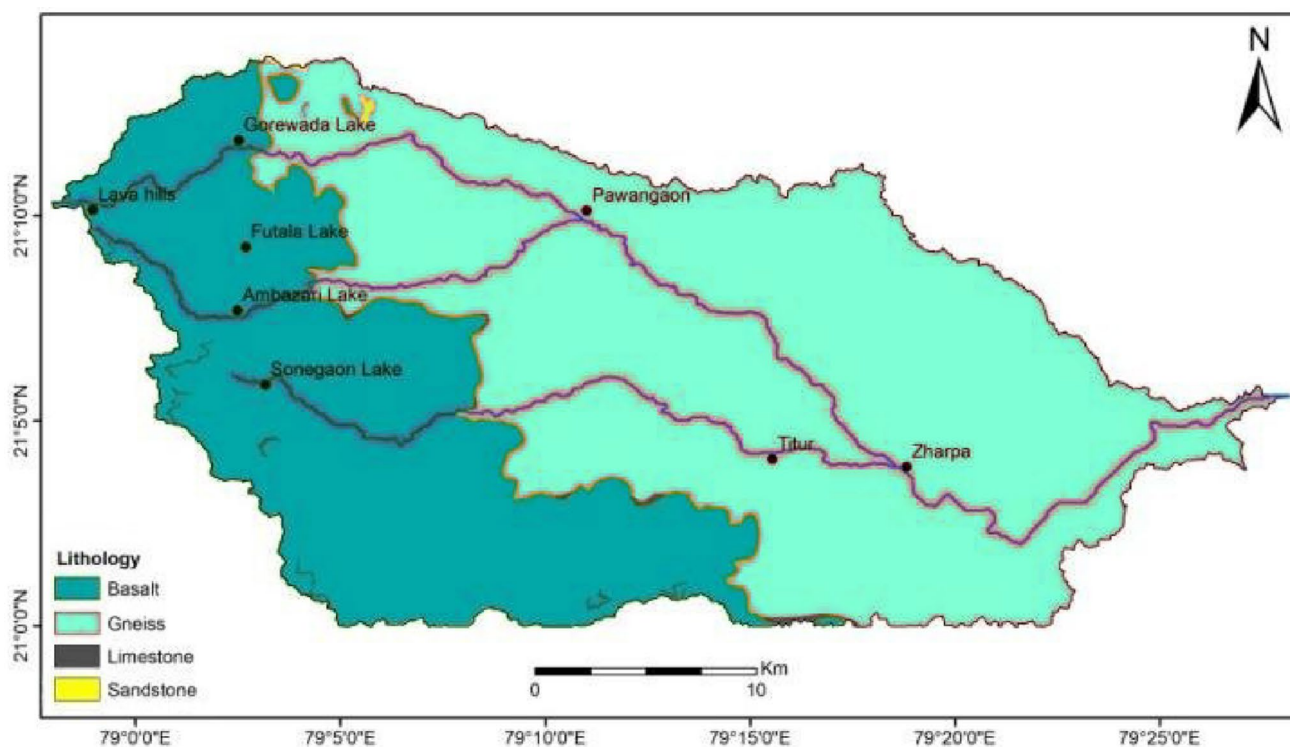


Fig. 1 Geologic map of the Nag River Basin study area, Nagpur, Maharashtra, India. (More et al. 2024)

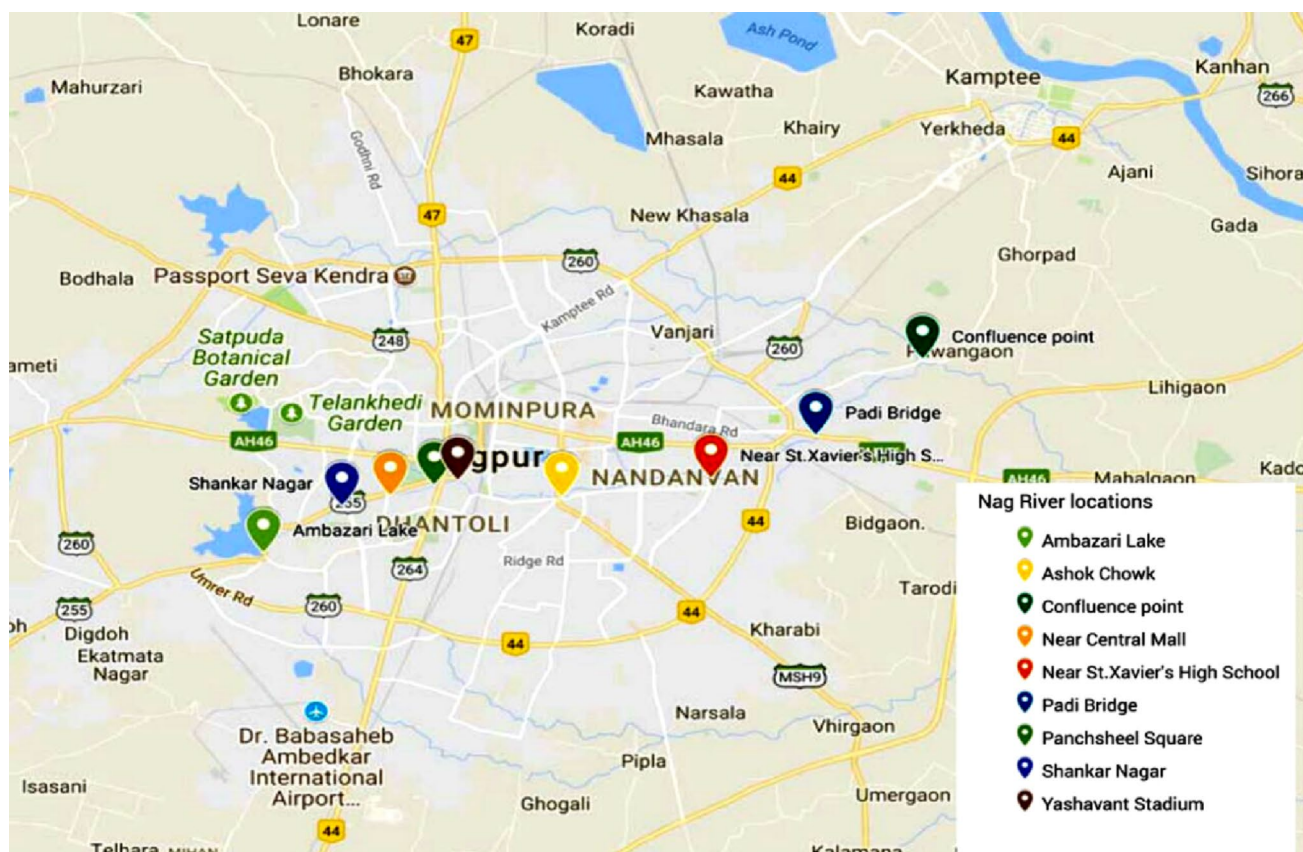


Fig. 2 Nag River sampling locations

Table 1 Systematic Labeling, Co-ordinates, and dominant land use context of sampling location

Sr. No.	Label	Location	Coordinates	Covering area	Dominant land use/ Intervention
1	S1	Ambazari Overflow	21°07'458'' N and 79°02'625'' E	Radius of 25 m	Baseline/ Recreational
2	S2	Shankar Nagar	21°07.461'N and 079°02.661'E	Radius of 25 m	Residential/ Commercial Runoff
3	S3	Central Mall	21°08.104'N and 079°03.664' E	Radius of 25 m	Commercial/ Industrial Effluent
4	S4	Panchsheel Square	21°08.337'N and 079°04.769'E	Radius of 25 m	Residential/ Small Scale Industry
5	S5	Yashwant Stadium	21°08.404' N and 079°04.925' E	Radius of 25 m	High-Density Residential Sewage Input
6	S6	Ashok Chowk	21°08.217'N and 079°06.267' E	Radius of 25 m	High-Density Residential Sewage Input
7	S7	Near St. Xavier School	21°08.441'N and 079°08.052'E	Radius of 25 m	Mixed Urban Runoff/Slums
8	S8	Pardi Bridge	21°08'454'' N and 79°10'562'' E	Radius of 25 m	Slum Waste-water/Minor Industry
9	S9	Confluence Point	21°08'104'' N and 79°03'664'' E	Radius of 25 m	Downstream Accumulation/ Final Discharge

Table 2 Methods for determining the water quality parameters

Phase	Activity	Outcome
I. Data Acquisition	Systematic Sampling (9 Sites: S1-S9) along 17 km Urban Stretch	Representative Spatial Coverage of Pollution Inputs
II. Laboratory Analysis	20 Parameters (Physico-chemical, Heavy Metal, Microbiological) (APHA 22nd Edition)	Raw Data Set
III. Quality Control	Triplicate Analysis, Standards/Blanks; Ionic Balance Error (IBE) Validation	Credibility Check; IBE <±5%
IV. Indexing and Standardization	Weighted Arithmetic WQI Calculation; Data Normalization (Z-score)	Water Quality Status; Preparation for Multivariate Analysis
V. Multivariate Analysis	PCA (Varimax Rotation for Source Apportionment); CA (AHC, Ward's Method)	Identification of Dominant Pollution Factors (VFs); Spatial Grouping of Sites (Clusters C1-C4)
VI. Synthesis	WQI-PCA-CA Integrated Interpretation and Land Use Linkage	Data-Driven Policy Recommendations

major urban runoff channels, identified industrial effluent outfalls, and areas of high-density residential discharge. The stations are representative of the upstream, midstream, and downstream stretches. S1 (Ambazari Lake Overflow) serves as the upstream baseline, while S9 (Confluence Point) represents the terminal monitoring location of the 17 km urban corridor. The linear distance between consecutive sampling sites was approximately 1.5 to 3.0 km, enabling a high-resolution analysis of incremental pollution loading due to human interventions along the river.

Sample collection and analysis

Nine surface water samples were collected in February 2023 between the hours of 7 a.m. to 10 a.m. These samples have been taken in 1-liter plastic bottles and 300 ml Borosil glass jars from locations about ½ meters below the surface in the middle of the water body. The bottles were filled with Nag River water, with the stopper put solely inside the water. Table 2 lists all the parameters that were tested, along with the units and procedures used. At the sampling site, pH and DO were corrected immediately by a calibrated portable pH and DO meter. For the remaining physicochemical parameters, the taken samples were subsequently refrigerated at 4 °C until the analysis was finished, which was completed in 7 days as opposed to 2 days for microbiological analysis. As per APHA 22nd edition (2012), microbiological and physicochemical parameters and heavy metal concentrations in water samples were measured. To place the selected analytical techniques in context, several alternative methods commonly used in river water quality studies were considered. For heavy metal analysis, ICP-OES, ICP-MS, and atomic absorption spectroscopy (AAS) are the principal options; ICP-MS provides the lowest detection limits (ppt range) but at substantially higher capital and operating costs and with more stringent requirements for operator expertise, whereas AAS is economical but limited to sequential single-element measurements, making it inefficient for multi-element monitoring campaigns. ICP-OES offers a balance between sensitivity (ppb range), throughput, and cost, enabling simultaneous quantification of multiple metals and therefore is widely adopted in surface- and groundwater monitoring, which supports its use in this multi-metal Nag River study. For mercury, direct mercury analyzers avoid wet-chemical pre-digestion and cold-vapor generation, reducing contamination risk and total analysis time compared with traditional CV-AAS, while still achieving trace-level quantification appropriate for environmental waters. Nitrate and sulfate were determined by UV-Vis spectrophotometer, which is routinely recommended in standard water analysis compendia and provides adequate selectivity and sensitivity at the relatively high concentrations expected in urban river

systems, without the higher cost and complexity of chromatographic methods.

These established methodologies (e.g., ICP-OES for heavy metals) were adopted due to their recognized high precision and consistency in hydrochemical analysis, thereby giving credence to the study's findings, aligning with best practices in environmental monitoring. The rigorous Quality Assurance and Quality Control (QA/QC) procedure followed both in the field and laboratory is crucial for the reliability of the results. To avoid any contamination and error, every safety measure was taken before the analysis. Triplicate examinations of standard, blank, and reference materials, alongside the water samples, were performed to assess the accuracy and precision of the experimental findings.

An ionic balance error (IBE) calculation was performed to validate the hydrochemical data accuracy. The IBE was computed for each sample using the formula below (Omeka 2024):

$$IBE = \frac{\sum \text{cations (meq/L)} - \sum \text{anions (meq/L)}}{\sum \text{cations (meq/L)} + \sum \text{anions (meq/L)}}$$

The table below summarizes the IBE results across the 9 sampling sites:

Sampling site	Distance from upstream (km)	Ionic balance error (IBE, %)
Site 1	0	2.3
Site 2	1.5	1.8
Site 3	3.0	3.1
Site 4	4.5	2.9
Site 5	6.0	4.0
Site 6	7.5	3.5
Site 7	9.0	2.7
Site 8	10.5	1.9
Site 9	12.0	3.2

All IBE values are within the acceptable $\pm 5\%$ range, indicating a good charge balance and confirming the reliability of the hydrochemical data obtained during the study. This validates the quality of the analytical measurements and supports the credibility of the water quality assessment results.

The overall methodology

The overall methodology utilized an integrated, multi-phased approach to link raw chemical data into actionable spatial pollution sources.

Sr. No.	Water quality parameters	Units	Methods of determination
1.	Hydrogen ion concentration (pH)	pH units	pH meter
2.	Total Hardness (TH)	mg/l	Herner's method
3.	Total Dissolved Solid (TDS)	mg/l	Evaporation method
4.	Sulphate (SO_4^{2-})	mg/l	UV-Vis spectrophotometer
5.	Nitrate (NO_3^-)	mg/l	UV-Vis spectrophotometer
6.	Chloride (Cl^-)	mg/l	Titration by silver nitrate
7.	Dissolved Oxygen (DO)	mg/l	DO meter/Azide modification at 20 °C for 1st day
8.	Biological Oxygen demand (BOD)	mg/l	Azide modification at 20 °C for 5 days
9.	Cadmium (Cd)	mg/l	ICP-OES
10.	Chromium (Cr)	mg/l	ICP-OES
11.	Copper (Cu)	mg/l	ICP-OES
12.	Iron (Fe)	mg/l	ICP-OES
13.	Mercury (Hg)	mg/l	Mercury analyzer
14.	Manganese (Mn)	mg/l	ICP-OES
15.	Nickel (Ni)	mg/l	ICP-OES
16.	Lead (Pb)	mg/l	ICP-OES
17.	Zinc (Zn)	mg/l	ICP-OES
18.	E.coli (EC)	CFU/100 ml	MPN method
19.	Fecal coliform (FC)	CFU/100 ml	MPN method
20.	Total coliform (TC)	CFU/100 ml	MPN method

Water quality index (WQI)

The water quality index (WQI) is a unitless number that is combined with multiple water parameters to represent the overall quality of water by reducing large values into a single index number. WQI is a highly useful mathematical function for determining the current state of water quality (Sener et al. 2017). Data for all physicochemical parameters were collected, and with the help of Microsoft Excel 2007, the weighted arithmetic WQI was calculated. The most frequently used physicochemical parameters that help assess the river's current pollution state, namely pH, total hardness, total dissolved solids (TDS), sulfate, nitrate, chloride, dissolved oxygen (DO), and biological oxygen demand (BOD), were considered for determining the water quality index (WQI). The weighted arithmetic WQI value for each parameter was determined using a formula (Isaac et al. 2022).

$$WQI = \frac{\sum W_n Q_n}{\sum W_n}$$

W_n = unit weight factor of each parameter, $\sum W_n = 1$

$$W_n = \frac{K}{S_n}$$

K is calculated by following equation

Table 3 Categorization of water quality status using the WQI range

Class	WQI range	Water quality
A	0–25	Excellent
B	26–50	Good
C	51–75	Poor
D	76–100	Very poor
E	> 100	Unsuitable for drinking use

$$K = \sum \left(\frac{1}{S_n} \right)$$

Where, K = proportionality constant

S_n = Standard permissible value of the parameter.

$$Q_n = \frac{[V_n - V_o]}{[S_n - V_o]} \times 100$$

Where, Q_n = sub-index of each selected parameter.

S_n = Standard desirable value of the nth parameter.

V_n = Mean concentration value of the nth parameter.

V_o = Ideal values of the parameter in pure water ($V_o = 0$, except $pH = 7$ and $DO = 14.6 \text{ mg/l}$)

Table 3 illustrates the quality of the river waters depending on the WQI value (Brown et al. 1972).

Multivariate statistical analysis

Multivariate statistical data analysis was carried out using the Microsoft add-in program XLSTAT trial version by Addinsoft. Before analysis, all raw data underwent Z-score standardization to eliminate the influence of different measurement units and ensure that variables with larger numerical values did not artificially dominate the analysis.

PCA was employed for both dimensionality reduction (retaining components with eigenvalues greater than 1, based on the Kaiser criterion) and source apportionment. A new set of variables in PCA, referred to as principal components (PCs) (Sojka et al. 2008; Garizi et al. 2011),

is derived from a group of correlated variables. The use of Varimax rotation was specifically applied to maximize the factor loadings for a smaller number of variables, thus simplifying the physical interpretation of each pollution source (Le et al. 2017). Cluster analysis (CA) was performed using hierarchical agglomerative cluster analysis (AHC), applying Ward's method and the Euclidean distance, using the normalized data to group sampling sites based on similarities in pollution characteristics (Vega et al. 1998; Warsito et al. 2021; Li et al. 2021, 2022).

Results and discussion

Examination of river water samples

The analysis of 20 physicochemical and biological parameters demonstrated that many critical metrics significantly exceeded the acceptable limits set by the CPCB and BIS, confirming the Nag River's severe environmental degradation.

Concentration of hydrogen ion (pH)

The pH level serves as an indicator of whether an aqueous solution is either acidic or basic. Typically, pH doesn't directly impact human health, but an imbalance of acidity or alkaline levels in the body due to the intake of water can disturb the normal pH of the human body and potentially results into numerous health issues such as skin allergy, gastrointestinal problems, and metabolic alkalosis (Tajudin and Hamirudin 2020; Lal et al. 2022). The Nag River's pH measurements in February 2023 during the pre-monsoon are shown in Table 4. Notably, the river water in the examined

Table 4 Physicochemical parameters of Nag river

Label	Sampling location	pH	TDS	TH	Cl	NO ₃ ⁻	SO ₄ ²⁻	BOD	DO
S1	Ambazari	8.6±0.09	395±24.16	105.37±14.32	175.18±10.14	9.15±1.61	120.14±11.63	0.2±5.99	4.62±0.34
S2	Shankar Nagar	7.99±0.15	620.37±10.41	440.65±12.91	279.76±10.92	26.39±1.2	65.86±8.12	148.47±7.61	0.75±0.31
S3	Central Mall	8.16±0.12	692±29.4	389.84±16.12	260.4±5.46	23.26±2.37	102.74±10.2	189.65±3.44	0.39±0.42
S4	Panchsheel Square	8.19±0.11	695±26.65	392±11.39	285.74±11.08	20.2±1.88	115.32±10.47	190.19±3.98	0.43±0.26
S5	Yashwant Stadium	7.93±0.07	828±14.25	399.24±12.92	289.41±6.71	24.82±1.24	125.63±6.85	216.28±3.23	0.31±0.52
S6	Ashok Chowk	7.99±0.07	802.11±13.64	375.64±13.66	285.24±5.65	23.84±1.99	124.07±14.7	207.76±4.63	0.39±0.23
S7	St. Xavier High School	8.09±0.06	753.78±13.67	298.91±14.56	209.97±14.49	20.16±1.07	162.22±12.75	186.26±4.94	0.48±0.59
S8	Pardi Bridge	8.06±0.14	751.14±16.08	297.41±17.85	207.35±14.66	22.87±2.82	179.28±14.39	195.27±4.36	0.79±0.51
S9	Confluence Point	7.46±0.11	702.12±20.5	267.45±12.0	204.49±13.08	22.18±1.52	161.24±13.95	185.66±7.14	1.39±0.28
	Permissible limits CPCB mg/l	6.5–8.5	500	300	250	20	400	2	6

area exhibited alkaline properties ranging from 7.46 to 8.19. The acceptable pH range for drinking water according to BIS and CPCB is 6.5–8.5 (BIS 2012 and CPCB 2007). However, the elevated alkaline pH values may be attributed to the release of commercial and household wastewater as well as the presence of detergents.

Total dissolved solids (TDS), total hardness (TH)

TDS (total dissolved solids) determines the concentration of inorganic salts and organic molecules present in the water sample; The TDS value is inversely proportional to the quality of the water. The TDS of the water was determined to be beyond permissible boundaries due to the lower level of water in the river and the higher solids content. The high TDS values indicate pollution of river water attributable to intense anthropogenic actions such as uncontrolled discharge of industrial waste, municipal and residential garbage, and agricultural land runoff (Tyagi et al. 2020). Total hardness is the amount of Calcium and Magnesium concentrations found in water samples. It was observed that the total hardness of the water slightly exceeded the permissible limit, as depicted in Table 4.

BOD and DO

The levels of Biological Oxygen Demand (BOD) and Dissolved Oxygen (DO) are primary indicators of organic pollution and the river's ecological health. The overall high BOD values in this study imply substantial levels of organic contamination, primarily originating from untreated sewage (Singh et al. 2020). Critically, the DO values for nearly every test site, starting from S2 onwards, extended dangerously low, ranging from 0.31 to 4.62 mg/l (Table 4). This acute deoxygenation, coupled with extremely high BOD values (peaking at 216.28 mg/l at Yashwant Stadium, S5), signifies that the majority of the urban river stretch is biologically stressed and ecologically unsuitable for supporting most aquatic life. This condition represents an acute ecological collapse caused by the excessive microbial breakdown of uncontrolled organic matter inputs, confirming the river is functioning as an open sewer (Sharma et al. 2014). This observation aligns with similar findings in the heavily polluted Yamuna River (Sharma et al. 2020) and Ganga River (Sharma et al. 2016), where high BOD is directly attributed to pervasive untreated domestic sewage and industrial waste, confirming a common driver for pollution in major Indian urban waterways.

Detergents (Chloride, nitrate, sulphate)

The primary sources of key ions in water are atmospheric deposition, chemical weathering and anthropogenic pollution from factories and sewage that is dumped into rivers. Chloride is a sign of the pollution that sewage causes. Living things and plants both eat chloride in required amounts, but at spiked concentrations, it is detrimental to health. According to (Kumar and Puri 2012), the reported chloride levels, which ranged from 204.49 to 289.41 mg/l, are beyond than the permissible limit of 250 mg/l in most of the locations, as shown in Table 4. Due to gypsum's natural tendency to leach, sulfate can be found in water. The concentration of nitrate and sulfate in water rises as a result of runoff from fertilized soil, industrial waste, and home garbage. According to (BIS 2012), the recorded nitrate and sulfate amounts are within the allowed range. The total hardness of the water is directly influenced by magnesium, calcium, and sulfate. The results of this study indicate that total hardness is increased by the presence of the high concentration of these ions.

Heavy metal concentration by ICP-OES

The concentration of several metal ions, notably manganese (Mn), iron (Fe), and lead (Pb), was found to be over allowable limits in all sampling locations. The quantity of cadmium (Cd) exceeded the BIS-set permissible limit in three sites. Table 5 lists the various heavy metal concentrations that were tested and the acceptable limits (BIS 2012).

The high concentrations of Fe and Mn are mechanistically linked to the catastrophic low DO environment. In the acute deoxygenated and slightly alkaline conditions (pH 7.46 to 8.19) caused by heavy organic loading, naturally occurring iron and manganese oxides in the sediment (derived partially from the underlying Deccan Trap basalt geology) become chemically reduced and are subsequently mobilized into the water column, explaining the ubiquity of high Fe and Mn values (US EPA 2004). Conversely, the presence of specific heavy metals, such as mercury (Hg), found above allowable limits, and cadmium (Cd), are not typically linked to natural mobilization but rather to anthropogenic point sources. The detection of Hg and Cd suggests dedicated pollution streams from nearby manufacturing units, hospitals, or specialized industrial processes (as identified in VF2), which requires regulatory control statistically separate from the pervasive municipal sewage problem (Zhou et al. 2020).

Table 5 Heavy metal concentration of Nag river

Label	Sampling Location	Cd	Cr	Cu	Fe	Hg	Mn	Ni	Pb	Zn
S1	Ambazari	ND	0.021 ± 0.002	0.004 ± 0.001	1.947 ± 0.213	ND	0.294 ± 0.025	0.009 ± 0.001	0.073 ± 0.005	0.01 ± 0.001
S2	Shankar Nagar	0.091 ± 0.005	0.023 ± 0.003	0.015 ± 0.002	3.539 ± 0.312	0.002 ± 0.0002	0.526 ± 0.038	0.012 ± 0.002	0.093 ± 0.007	2.976 ± 0.254
S3	Central Mall	0.092 ± 0.006	0.018 ± 0.002	0.014 ± 0.001	3.654 ± 0.298	0.004 ± 0.0003	0.423 ± 0.041	0.023 ± 0.003	0.097 ± 0.006	3.384 ± 0.287
S4	Panchsheel Square	0.086 ± 0.004	0.025 ± 0.002	0.023 ± 0.002	3.632 ± 0.275	0.003 ± 0.0002	0.436 ± 0.035	0.021 ± 0.002	0.086 ± 0.005	3.276 ± 0.261
S5	Yashwant Stadium	ND	0.028 ± 0.003	0.026 ± 0.002	4.468 ± 0.352	0.004 ± 0.0003	0.759 ± 0.063	ND	0.145 ± 0.009	5.569 ± 0.312
S6	Ashok Chowk	ND	0.022 ± 0.002	0.025 ± 0.002	4.394 ± 0.341	0.003 ± 0.0002	0.756 ± 0.058	ND	0.129 ± 0.008	4.294 ± 0.298
S7	St. Xavier High School	ND	0.017 ± 0.001	ND	4.128 ± 0.324	0.001 ± 0.0001	0.492 ± 0.049	ND	0.053 ± 0.004	3.624 ± 0.287
S8	Pardi Bridge	ND	0.019 ± 0.002	ND	3.264 ± 0.276	0.001 ± 0.0001	0.375 ± 0.031	ND	0.053 ± 0.004	3.242 ± 0.276
S9	Confluence Point	ND	0.019 ± 0.002	ND	2.159 ± 0.198	ND	0.266 ± 0.024	ND	0.046 ± 0.003	2.111 ± 0.214
	Acceptable limits BIS	0.003	0.05	0.05	0.3	0.001	0.1	0.02	0.01	5

ND not detected

Table 6 Microbiological parameters of Nag river

Label	Sampling location	E. coli	Fecal coliform	Total coliform
S1	Ambazari	1400 ± 250	2700 ± 500	6600 ± 900
S2	Shankar Nagar	7,300,000 ± 350,000	9,500,000 ± 470,000	10,600,000 ± 520,000
S3	Central Mall	6,900,000 ± 330,000	9,700,000 ± 480,000	14,600,000 ± 690,000
S4	Panchsheel Square	7,700,000 ± 370,000	13,600,000 ± 680,000	15,600,000 ± 780,000
S5	Yashwant Stadium	9,700,000 ± 480,000	18,500,000 ± 930,000	29,500,000 ± 1,470,000
S6	Ashok Chowk	8,500,000 ± 420,000	18,300,000 ± 910,000	28,500,000 ± 1,420,000
S7	St. Xavier High School	8,900,000 ± 440,000	7,600,000 ± 380,000	17,500,000 ± 870,000
S8	Pardi Bridge	5,900,000 ± 290,000	6,500,000 ± 320,000	17,000,000 ± 850,000
S9	Confluence Point	5,600,000 ± 280,000	7,300,000 ± 360,000	15,400,000 ± 770,000
	Acceptable limits BIS	0	0	0

Microbial analysis

Table 6 lists the major microbiological metrics as well as the (BIS 2012), which provides acceptable limits. Table 6 clearly shows that the values for E. coli, fecal coliform, and total coliform were significantly greater than the recommended value provided by (BIS 2012). Directly dumping untreated residential sewage, animal waste, and human waste into rivers can cause bacterial contamination. It was discovered that feces were present at every sampling site. According to the findings of the present investigation, Nehme et al. (2018) observations from their study on the Litani River suggest that fecal coliform growth has been impacted by high concentration of some specific heavy metals.

Water quality index (WQI)

The weighted arithmetic WQI effectively translates the complex analytical data into a single, intuitive number. The results confirmed a catastrophic degradation of water quality along the urban stretch. The outcomes of a few chosen water quality parameters are necessary for computing WQI. Table 7 represents the unit weights (W_i) of the water quality parameters for WQI determination. The calculated WQI values for the Nag River showed a severe range, varying from 47.05 at the upstream S1 site to a staggering peak of 6440.38 at S5. The S1 (Ambazari) location is classified as 'Good,' serving as the required environmental baseline. However, starting immediately from S2 (Shankar Nagar), the WQI jumps steeply, classifying the water as 'Unfit for all uses' according to the established categorization. The

Table 7 CPCB standards, unit weights and ideal value of parameters applied for WQI determination

Water parameters	CPCB permissible value (Sn)	Unit weight (Wi)	Ideal value (Vo)
pH	6.5–8.5	0.139039	7
Total Hardness	300 mg/l	0.003939	0
TDS	500 mg/l	0.002364	0
Sulfate	400 mg/l	0.002955	0
Nitrate	20 mg/l	0.059091	0
Chloride	250 mg/l	0.004727	0
DO	6 mg/l	0.196971	14.6
BOD	2 mg/l	0.590914	0

Table 8 WQI value and water quality status of sampling location (S1–S9)

Label	Sampling location	WQI value	Water quality status
S1	Ambazari	47.05	Good
S2	Shankar Nagar	4436.79	Unfit for all uses
S3	Central Mall	5654.92	Unfit for all uses
S4	Panchsheel Square	5670.22	Unfit for all uses
S5	Yashwant Stadium	6440.38	Unfit for all uses
S6	Ashok Chowk	6188.68	Unfit for all uses
S7	St. Xavier High School	5553.21	Unfit for all uses
S8	Pardi Bridge	5818.87	Unfit for all uses
S9	Confluence Point	5528.07	Unfit for all uses

calculated weighted arithmetic WQI for the Nag River is shown in Table 8.

Based on the WQI values obtained from this study, it is revealed that the water quality of the Nag River throughout its total length is very low and unsuitable for any usage throughout the pre-monsoon season, in accordance with the particulars given by Brown et al. (1972), as mentioned in Table 3.

For strengthening methodological framework, the current research offers clear cut threshold values for WQI classification. According to CPCB norms, $WQI > 100$ is water with heavy pollution, $WQI 76–100$ is poor water quality, and $WQI < 50$ is acceptable conditions. The grading allows direct comparison with similar studies and ensures consistency in interpretation of water quality status.

Referring to recommended WQI assessment practices, such as those applied to the Godavari River (Kharake and Raut 2021), the ideal WQI value for pure or near-potable water approaches zero. WQI values exceeding 6000, as seen at S5 and S6, represent an exceptional state of contamination nearly 130 times worse than the river's clean source point S1 signifying that the pollution load is overwhelming the river's entire ecosystem. The spatial trend confirms that municipal sewage and garbage inputs significantly contribute to pollution and drastically increase the WQI immediately upon entering the urban reach (Chaurasia et al. 2018).

Table 9 Strength of correlation to understand correlation matrix

Correlation Strength	Positive	Negative
Very Strong	$r = 0.9$ to 1	$r = -0.9$ to -1
Strong	$r = 0.5$ to 0.9	$r = -0.5$ to -0.9
Weak	$r = 0.1$ to 0.5	$r = -0.1$ to -0.5
Uncorrelated	$r = 0$ to 0.1	$r = 0$ to -0.1

The application of WQI in this study provides a quantitative means by which to measure trends in river water quality and pollution sources. The method developed enables policymakers to enact targeted mitigation measures, including tighter wastewater treatment policy, industrial effluent control, and community-based pollution mitigation. By providing a replicable model, this study makes a contribution to improved water quality governance in urbanized river systems.

Correlation of water parameters

Pearson's correlation matrix was utilized to quantitatively determine the relationship between the 20 water quality parameters. Strong correlations ($r > \pm 0.5$) were highlighted to confirm shared sources of pollution.

The correlation matrix provides rigorous confirmation of shared pollution vectors. A very strong positive correlation ($r = 0.96$) between BOD and TDS, alongside a very strong negative correlation with DO ($r = -0.96$), demonstrates that the primary pollution plume is a unified input of untreated municipal sewage. This sewage is rich in organic matter (BOD), dissolved salts (TDS, Chloride), fecal bacteria, and associated trace metals (e.g., Zn from corrosion and waste). Furthermore, the strong positive correlation identified between Total Hardness (TH) and Chloride (Cl^-) ($r = 0.90$) confirms that the elevated TH observed is not merely a consequence of natural geological background but is predominantly driven by the dissolution of chloride salts originating from domestic inputs (detergents and wastewater).

Specifically, a robust positive correlation ($r = 0.53$) emerged between dissolved oxygen (DO) and pH, indicating that pH might affect the quantity of oxygen dissolved in water leading to lower microbial counts. Total dissolved solids (TDS) exhibited a robust and positive correlation with total hardness ($r = 0.66$), chloride ($r = 0.53$), nitrate ($r = 0.78$), fecal coliform ($r = 0.78$), iron (Fe) ($r = 0.76$), mercury (Hg) ($r = 0.52$), and manganese (Mn) ($r = 0.62$). Singh et al. (2014) reported similar findings, attributing causes to public bowel movements, improper solid waste disposal, and sewage runoff (Tables 9 and 10).

Table 10 Pearson's correlation matrix between different water parameters (Strong correlations $|r| \geq 0.5$ are highlighted)

	pH	TDS	TH	Cl ⁻	NO ₃ ⁻	SO ₄ ²⁻	BOD	DO	TC	FC	EC	Cd	Cr	Cu	Fe	Hg	Mn	Ni	Pb	Zn
pH	1																			
TDS	-0.61	1																		
TH	-0.37	0.66	1																	
Cl⁻	-0.18	0.53	0.90	1																
NO₃⁻	-0.67	0.78	0.89	0.68	1															
SO₄²⁻	-0.27	0.32	-0.44	-0.59	-0.15	1														
BOD	-0.66	0.96	0.76	0.58	0.85	0.22	1													
DO	0.53	-0.89	-0.88	-0.69	-0.89	-0.01	-0.96	1												
TC	-0.51	0.94	0.60	0.61	0.68	0.20	0.84	-0.76	1											
FC	-0.36	0.78	0.77	0.88	0.67	-0.20	0.74	-0.74	0.89	1										
EC	-0.49	0.90	0.84	0.72	0.82	-0.02	0.90	-0.94	0.84	0.82	1									
Cd	0.15	-0.14	0.57	0.52	0.29	-0.74	0.08	-0.29	-0.25	0.10	0.15	1								
Cr	0.02	0.15	0.39	0.66	0.18	-0.43	0.10	-0.12	0.36	0.61	0.27	0.13	1							
Cu	0.05	0.35	0.66	0.92	0.38	-0.59	0.34	-0.41	0.55	0.85	0.50	0.36	0.79	1						
Fe	-0.08	0.76	0.75	0.74	0.63	-0.15	0.70	-0.79	0.79	0.82	0.89	0.12	0.33	0.64	1					
Hg	0.02	0.52	0.78	0.88	0.54	-0.47	0.55	-0.64	0.60	0.80	0.65	0.48	0.51	0.85	0.76	1				
Mn	-0.10	0.62	0.62	0.75	0.52	-0.27	0.49	-0.54	0.80	0.87	0.71	-0.08	0.57	0.77	0.87	0.70	1			
Ni	0.42	-0.36	0.26	0.31	-0.09	-0.64	-0.17	0.00	-0.42	-0.08	-0.13	0.90	0.06	0.29	-0.07	0.41	-0.25	1		
Pb	0.07	0.33	0.54	0.79	0.35	-0.53	0.26	-0.30	0.59	0.79	0.44	0.13	0.73	0.91	0.64	0.82	0.86	0.08	1	
Zn	-0.39	0.92	0.79	0.74	0.78	0.01	0.87	-0.87	0.92	0.88	0.94	0.02	0.40	0.60	0.92	0.75	0.82	-0.20	0.62	1

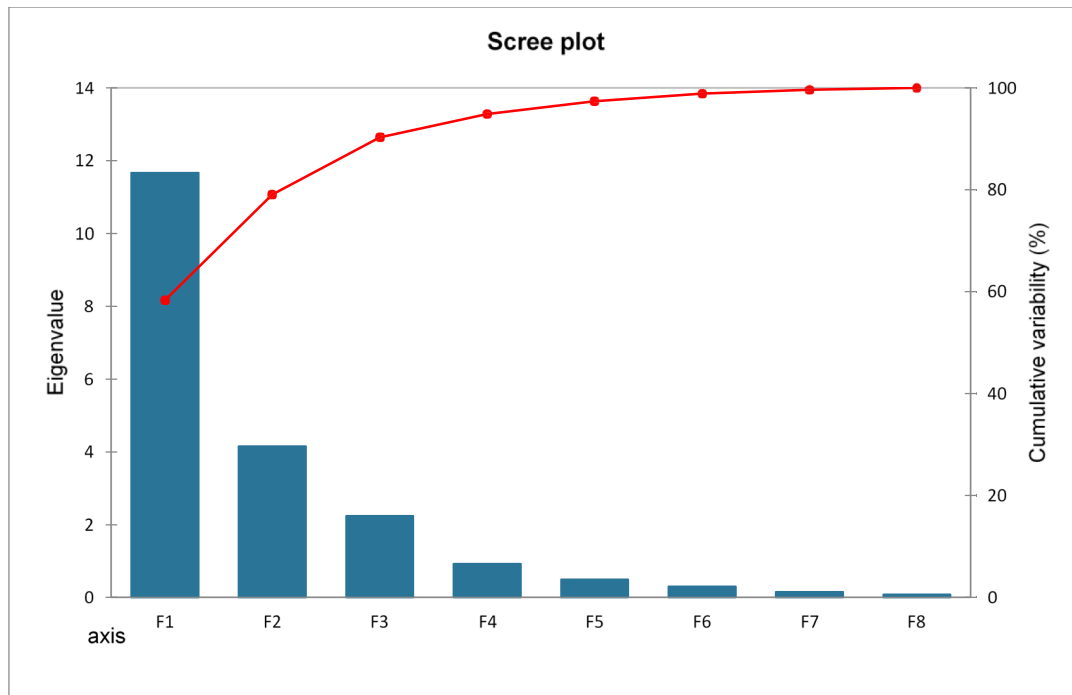


Fig. 3 The eigenvalue scree diagram for each component

Table 11 Factor loadings of the experimental data on the five significant PCs

	PC1	PC2	PC3	PC4	PC5
pH	−0.385	−0.603	−0.297	0.602	−0.019
TDS	0.833	0.530	0.063	0.090	0.109
TH	0.899	−0.174	0.366	−0.077	−0.109
Cl [−]	0.899	−0.406	0.039	−0.116	0.016
NO ₃ [−]	0.819	0.215	0.385	−0.228	−0.205
SO ₄ ^{2−}	−0.232	0.879	−0.107	0.211	0.330
BOD	0.830	0.438	0.301	0.027	0.163
DO	−0.872	−0.249	−0.399	−0.110	−0.011
TC	0.872	0.397	−0.244	0.021	0.098
FC	0.954	−0.023	−0.203	−0.085	0.127
EC	0.920	0.243	0.157	0.103	−0.074
Cd	0.227	−0.707	0.667	0.002	0.012
Cr	0.506	−0.433	−0.476	−0.384	0.228
Cu	0.765	−0.531	−0.287	−0.091	0.146
Fe	0.891	0.026	−0.105	0.386	−0.176
Hg	0.837	−0.415	0.004	0.209	0.152
Mn	0.840	−0.060	−0.464	0.081	−0.259
Ni	−0.017	−0.801	0.506	0.173	0.256
Pb	0.718	−0.433	−0.486	−0.049	−0.048
Zn	0.952	0.216	−0.064	0.136	0.003
Variability (%)	58.319	20.755	11.187	4.616	2.474
Eigenvalue	11.664	4.151	2.237	0.923	0.495
Cumulative (%)	58.319	79.074	90.261	94.877	97.351

Principal component analysis (PCA)

PCA was applied to the Z-score standardized data to identify the dominant, latent factors driving water quality variability and to perform source apportionment. Based on the Kaiser criterion (eigenvalue > 1), three Varifactors (VFs) were retained after Varimax rotation, collectively explaining **87.935%** of the total cumulative variance (Omeka et al. 2024; Ucheana et al. 2024). Principal component analysis (PCA) assists to find patterns in data and representing the data in such a format that facilitates the identification of differences and similarities among variables. It proves particularly useful in identifying and graphically representing the ecological components of environmental systems (Isaac et al. 2022). The corresponding eigenvalue for each principal component is displayed in the Scree plot seen in Fig. 3. The fundamental data co-relation of the study is evaluated by this technique (Han et al. 2009; Mohamed et al. 2015). The scree plot shows a noticeable alteration in slope that can be observed after the fifth PC.

Table 11 illustrates the factor loadings for the five principal components. Factor loadings greater than 0.5 were considered to be significant, as per (Dutta et al. 2018), indicating strong correlation with specific pollution indicators.

Focusing on PC1, it contributes to an eigenvalue of 11.664, explaining 58.319% of the variance. TDS, TH, chloride, nitrate, BOD, fecal coliform, total coliform, E. coli, and Fe, Cr, Cu, Hg, Mn, and Zn make substantial contributions to PC1. Conversely, pH, sulfate, DO, and Ni exhibit

negative participation in PC1. Moving to PC2, with an eigenvalue of 4.151, it accounts for 20.755% of the variance, with substantive positive loadings from TDS and sulfate and negative loadings from pH, Cd, Cu, and Ni. The principal component, PC3, with an eigenvalue of 2.237 and a variance of 11.187%, is primarily influenced by nickel and cadmium. PC4 explains a variance of 4.616% with an eigenvalue of 0.923, where pH is the sole significant positive loading. Lastly, PC5, with a variance of 2.474% and an eigenvalue of 0.495, displays no significant contribution from any parameters. Due to the extremely random nature of all the principal components, the physicochemical and biological interpretations are hampered. Therefore, by enhancing the more significant variables, a significant rotation of the principal components was performed for more simple and pertinent representation. According to (Singh et al. 2013), rotation changes the variance that each factor explains. Table 12 displays the factor loadings for the varimax rotated components, also referred to as varifactors.

Varifactor interpretation (Source apportionment)

Varifactor 1 (VF1): Anthropogenic Organic Load Factor (44.088% variance) VF1 is the dominant factor, characterized by extreme loadings from BOD (0.957), DO (-0.968), Electrical Conductivity (0.929), Total Dissolved Solids (0.941), Total Coliform (0.807), Nitrate (0.821), Iron

(0.811), and Zinc (0.885). This factor quantifies the overwhelming influence of untreated municipal sewage entering the river, confirming the source is a unified plume of organic matter, dissolved salts, fecal bacteria, and associated metals. The negative loading of DO confirms that the large organic load drives acute deoxygenation (Vega et al. 1998).

Varifactor 2 (VF2): Specialized Industrial Factor (16.666% variance) VF2 is uniquely defined by overwhelming positive loadings from Cadmium (0.987) and Nickel (0.948). The statistical isolation of these two heavy metals from the general sewage factor (VF1) is the critical outcome of this multivariate analysis. This separation provides quantitative evidence of dedicated, localized point-source contamination streams, highly typical of specialized industrial activities (e.g., electroplating or metal processing) that discharge non-compliant effluents.

Varifactor 3 (VF3): Mixed Inorganic and Generalized Heavy Metal Factor (27.182% variance) VF3 is characterized by strong loadings from Chloride (0.701), Chromium (0.911), Copper (0.898), Lead (0.910), and Manganese (0.722). This factor represents a secondary, diffuse pollution vector, encompassing inorganic salt buildup from general urban runoff and a mix of heavy metals associated with generalized urban and commercial waste discharge, distinct from the acute sewage and specialized industrial inputs.

Notably, a study on the Nampong and Songkhram Rivers in Thailand (Mohamed et al. 2015) produced comparable results. These findings provide a scientific base for policy recommendations, including increased industrial discharge regulation, targeted wastewater treatment initiatives, and improved agricultural runoff control. The PCA methodology applied in this study is a replicable blueprint for similar urban river studies, enabling consistency in pollution classification and abatement strategies. The results from PCA demonstrate its effectiveness in classifying pollution sources and assessing water quality deterioration in the Nag River, making it a valuable tool for environmental monitoring and remediation planning.

Figure 4 displays factor scores associated with specific pollution types. The Nag River's source, Ambazari (S1), exhibits a notable high pH value and remarkably low levels of all pollutants, serving as an indicator of exceptional water purity. This observation is also corroborated by the analysis of the Water Quality Index (WQI). Industrial pollution, specifically Cd and Ni discharge, is predominant in Shankar Nagar (S2), Central Mall (S3), and Panchsheel Square (S4). In contrast, St. Xavier (S7), Yashwant Stadium (S5), Ashok Chowk (S6), Central Mall (S3), and Pardi Bridge (S8) are primary discharge locations for various pollutants such as organic, heavy metal, fecal, or mixed types. Heavy metal pollution is particularly concentrated in Panchsheel Square (S4), Ashok Chowk (S6), and Yashwant Stadium (S5).

Table 12 Factor loading after varimax rotation (Loadings > 0.5 in bold italics)

	Varifactor 1	Varifactor 2	Varifactor 3	Varifactor 4	Varifactor 5
pH	-0.496	0.215	0.099	0.812	0.019
TDS	0.941	-0.246	0.156	-0.123	-0.116
TH	0.760	0.453	0.362	-0.189	0.201
Cl ⁻	0.555	0.411	0.701	-0.084	0.113
NO ₃ ⁻	0.821	0.161	0.146	-0.418	0.253
SO ₄ ²⁻	0.186	-0.700	-0.499	0.016	-0.467
BOD	0.957	-0.016	0.077	-0.228	-0.155
DO	-0.968	-0.180	-0.089	0.130	-0.014
TC	0.807	-0.354	0.447	-0.071	-0.077
FC	0.689	-0.013	0.701	-0.087	-0.039
EC	0.929	0.019	0.258	-0.067	0.105
Cd	0.103	0.987	0.059	-0.033	0.087
Cr	-0.008	0.057	0.911	-0.163	-0.098
Cu	0.300	0.282	0.898	0.064	-0.009
Fe	0.811	-0.003	0.419	0.327	0.211
Hg	0.573	0.410	0.624	0.227	-0.049
Mn	0.546	-0.203	0.722	0.174	0.326
Ni	-0.143	0.948	0.055	0.200	-0.173
Pb	0.252	0.056	0.910	0.148	0.165
Zn	0.885	-0.099	0.425	0.027	0.035
Variability (%)	44.088	16.666	27.182	6.310	3.106
Cumulative (%)	44.088	60.753	87.935	94.245	97.351

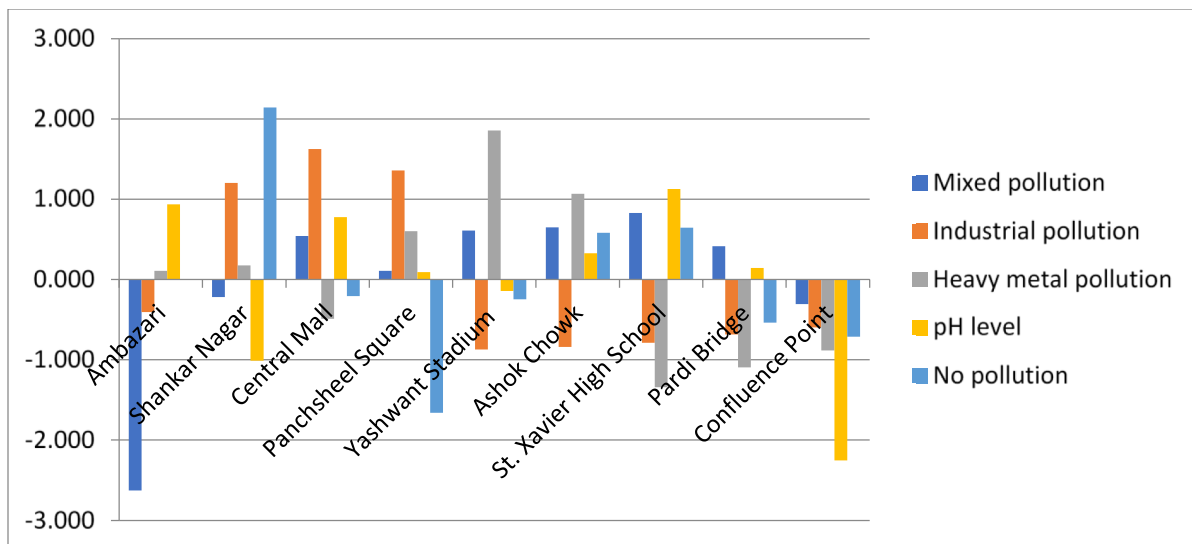


Fig. 4 Factor score graph of the different sampling location of Nag River

Table 13 Factor score after varimax rotation

	FS1	FS2	FS3	FS4	FS5
Ambazari	-2.627	-0.406	0.109	0.937	-0.004
Shankar Nagar	-0.220	1.203	0.177	-1.009	2.142
Central Mall	0.542	1.625	-0.491	0.778	-0.207
Panchsheel Square	0.109	1.356	0.600	0.091	-1.661
Yashwant Stadium	0.610	-0.873	1.854	-0.141	-0.245
Ashok Chowk	0.648	-0.839	1.069	0.326	0.580
St. Xavier High School	0.828	-0.788	-1.343	1.128	0.646
Pardi Bridge	0.416	-0.683	-1.093	0.142	-0.539
Confluence Point	-0.306	-0.596	-0.882	-2.252	-0.711

Notably, Confluence Point (S9) shows no contamination of any kind (Table 13).

Cluster analysis (CA)

Hierarchical Agglomerative Clustering (AHC) grouped the nine systematically labeled sampling sites (S1–S9) based on their pollution similarity, effectively providing a geographical map of contamination patterns. The dendrogram reveals four distinct spatial clusters in Fig. 5.

Spatial clustering and land use linkage

- **Cluster C3 (Control):** Comprising only S1 (Ambazari). This cluster is the upstream baseline, characterized by minimal contamination and the highest pH values.
- **Cluster C2 (Industrial Hotspots):** Formed by S2 (Shankar Nagar), S3 (Central Mall), and S4 (Panchsheel Square). This initial urban segment is heavily influenced by the Specialized Industrial Factor (VF2: Ni and Cd), indicating that these sites, which are proximal

to commercial and small-scale industrial activities, require the highest priority for industrial compliance enforcement.

- **Cluster C1 (Peak Organic Contamination):** Comprising S5 (Yashwant Stadium) and S6 (Ashok Chowk). This cluster exhibits the maximum WQI values (up to 6440.38) and is overwhelmingly driven by the Anthropogenic Organic Load Factor (VF1: BOD, EC, Zn). These sites are located in high-density residential sectors and reflect the acute failure of the municipal sewer network.
- **Cluster C4 (Downstream Mixed Accumulation):** Including S7 (St. Xavier), S8 (Pardi Bridge), and S9 (Confluence Point). This downstream section shows moderate levels of mixed-type pollution (VF1 and VF3 persistence) and the accumulation of contaminants from upstream sources, receiving wastewater from homes, slums, and minor industrial facilities.

By integrating cluster analysis with spatial pollution analysis, this study provides a foundation for informed decision-making in urban water management. Future research should build on this framework by incorporating real-time pollution monitoring and predictive modeling to enhance the effectiveness of pollution mitigation strategies.

Study limitations

While this work is informative in terms of Nag River water quality, there are a few limitations that must be considered. One is that this work is a one-season analysis that would not account for all seasonality in water quality. In the future, more detailed analysis would be executed using a

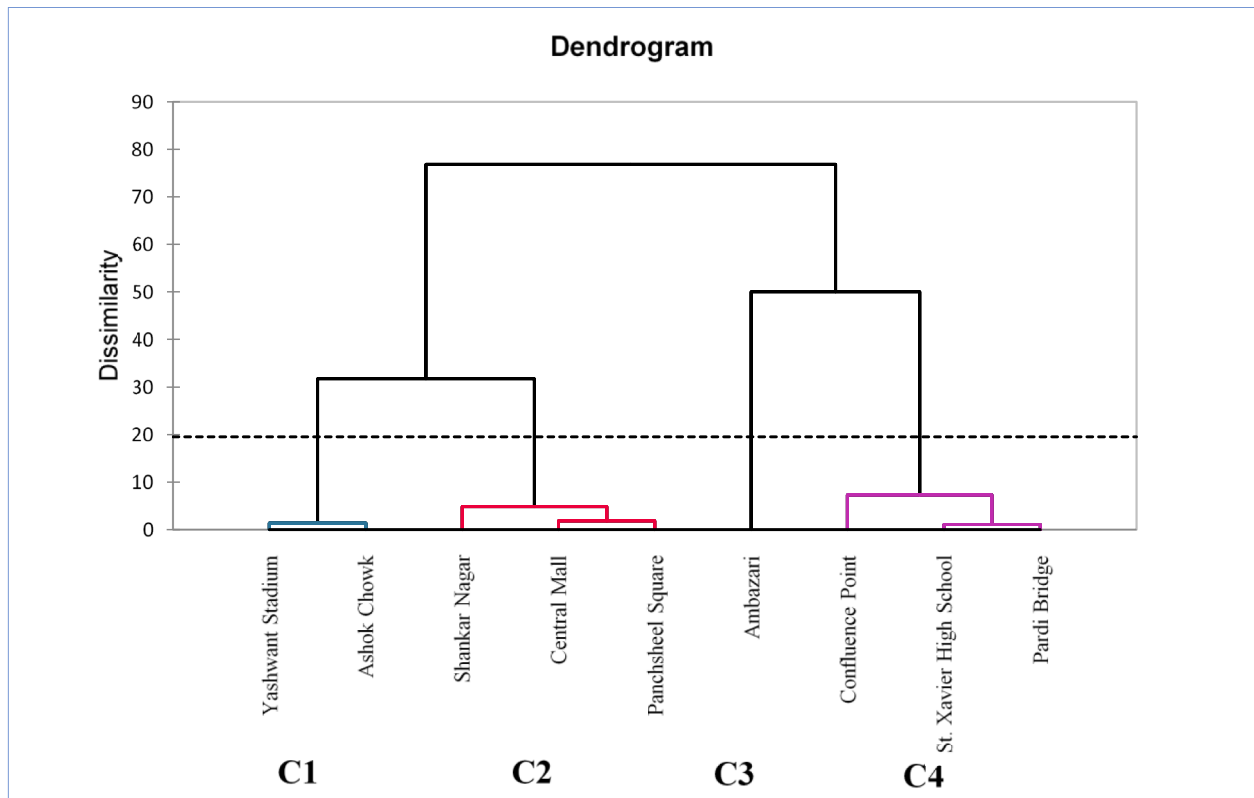


Fig. 5 Dendrogram created using agglomerative hierarchical clustering for sampling sites

multi-seasonal approach. Secondly, although multivariate statistics were employed, more advanced modeling tools such as machine learning would be helpful in enhancing predictive capability. Third, logistically, a subset of sites was sampled; more sites would be sampled to obtain a more detailed analysis of pollutant distribution.

Conclusion

The study successfully employed an integrated WQI-PCA-CA framework to provide a high-resolution assessment of the Nag River water quality along its 17 km urban stretch. The findings conclusively demonstrate that the river is in a state of severe environmental crisis. The calculated WQI values, ranging up to an exceptional 6440.38, clearly quantify the unsuitability of the water for virtually all uses.

The study concluded that the catastrophic degradation of the Nag River is a direct consequence of unchecked urbanization and rapid, non-compliant industrialization, evidenced by the extreme loading of untreated municipal wastewater and specialized industrial effluents. The multivariate statistical analysis successfully decomposed this complex contamination into two principal, manageable factors: Varifactor 1 (VF1, 44.088% variance) was identified as the primary Anthropogenic Organic Load Factor (municipal

sewage and high BOD); and Varifactor 2 (VF2, 16.666% variance) was defined as the Specialized Industrial Factor, uniquely driven by high concentrations of Nickel and Cadmium.

The integrated framework offers a competitive advantage over conventional assessments by transforming abstract statistical data into spatially defined, actionable intelligence. Cluster Analysis mapped these factors directly to land-use impacts: the C1 sites (Yashwant Stadium, Ashok Square) are the priority zones for large-scale municipal sewage treatment (VF1 mitigation), while the C2 sites (Shankar Nagar, Central Mall) are the priority zones requiring immediate, strict industrial regulation enforcement (VF2 mitigation).

Impacts of Human Interventions and Proposed Strategic Policy Interventions:

1. Mitigation of Organic Load (VF1): The pervasive discharge of untreated sewage from high-density residential and slum areas, particularly at the C1 and C4 clusters, necessitates immediate implementation of proper drainage systems and construction of tertiary municipal wastewater treatment plants to eliminate organic and microbial load.
2. Control of Specialized Industrial Effluents (VF2): The statistically isolated presence of Ni and Cd at the C2 sites demands a focused regulatory policy, including

mandatory pre-treatment systems for industrial units to ensure zero-discharge of specialized heavy metals into the river.

3. Societal and General Measures: The widespread practice of open defecation and solid trash dumping contributes significantly to the problem, requiring immediate public health awareness campaigns, provision of sanitation infrastructure, and the enforcement of comprehensive solid waste management policies along the riverbanks.

In conclusion, this research provides a replicable geospatial model for resource allocation and environmental policy development, underscoring the imperative need for immediate, data-informed action to achieve sustainable water quality restoration in the Nag River and offering a crucial blueprint for similar urban river rejuvenation projects globally.

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Ethical approval We declare herein that our paper is original and unpublished elsewhere and that this manuscript complies to the Ethical Rules applicable for this journal.

Consent to participate All of the authors consent to participate in this research work.

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